MEC6602E Transonic Aerodynamics

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Install Tecplot for Homework 2

- `https://fichiers.meca.polymtl.ca/Tecplot_Academic_Suite/Tecplot360/2023R1/`
- You will need to be logged on the Polytechnique Montréal VPN to have access
- You will need to send me:
  - Your name
  - Name of your computer
  - Your username on this computer
- Wait for your computer to be added, then set up the network licence:
  `https://fichiers.meca.polymtl.ca/Tecplot_Academic_Suite/License_File/`

2D Euler equations

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} + \frac{\partial (\rho w)}{\partial z} = 0$$

Continuity equation

$$\frac{\partial (\rho u)}{\partial t} + \nabla \cdot (\rho u \vec{V}) + \nabla p = 0$$
$$\frac{\partial (\rho v)}{\partial t} + \nabla \cdot (\rho v \vec{V}) + \nabla p = 0$$

Momentum equations

$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\rho E \vec{V}) + \nabla \cdot (p \vec{V}) = 0$$

Energy equation

Integral form of continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0$$

Integrate on control volumes Ω

$$\int_\Omega \frac{\partial \rho}{\partial t} \mathrm{d}\Omega + \int_\Omega \nabla \cdot (\rho \vec{V}) \mathrm{d}\Omega = 0$$

Divergence theorem: $\int_V \nabla \cdot \vec{F} \, \mathrm{d}V = \oint_S \vec{F} \cdot \mathrm{d}\vec{S}$

$$\boxed{\int_\Omega \frac{\partial \rho}{\partial t} \mathrm{d}\Omega + \oint_S \rho \vec{V} \cdot \mathrm{d}\vec{S} = 0}$$

Change of mass in the volume is the flux of mass through the boundaries.

Integral form of momentum equation

$$\frac{\partial \rho u}{\partial t} + \nabla \cdot (\rho u \vec{V}) + \nabla p = 0$$

$$\int_\Omega \frac{\partial \rho u}{\partial t} \mathrm{d}\Omega + \int_\Omega \nabla \cdot (\rho u \vec{V}) \mathrm{d}\Omega + \int_\Omega \nabla p \, \mathrm{d}\Omega = 0$$

$$\int_\Omega \frac{\partial \rho u}{\partial t} \mathrm{d}\Omega + \oint_S (\rho u \vec{V} + p) \cdot \mathrm{d}\vec{S} = 0$$

Integral form of energy equation

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot (\rho E \vec{V}) + \nabla \cdot (p \vec{V}) = 0$$

$$\int_\Omega \frac{\partial \rho E}{\partial t} \mathrm{d}\Omega + \int_\Omega \nabla \cdot (\rho E \vec{V}) \mathrm{d}\Omega + \int_\Omega \nabla \cdot (p \vec{V}) \mathrm{d}\Omega = 0$$

$$\int_\Omega \frac{\partial \rho E}{\partial t} \mathrm{d}\Omega + \oint_S (\rho E \vec{V} + p \vec{V}) \cdot \mathrm{d}\vec{S} = 0$$

Vector form

$$\mathrm{d}\vec{S} = \mathrm{d}S\,\vec{n}$$

$$\int_\Omega \frac{\partial \vec{W}}{\partial t} + \oint_S \vec{F}_c\,\mathrm{d}S = 0$$

$$\vec{W} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \cancel{\rho w} \\ \rho E \end{pmatrix}_{\text{3D}} \qquad \vec{F}_c = \begin{pmatrix} \rho V_c \\ \rho u V_c + n_x p \\ \rho v V_c + n_y p \\ \cancel{\rho w V_c + n_z p} \\ \rho E V_c + p V_c \end{pmatrix}_{\text{3D}}$$

Contravariant velocity: $V_c = \vec{V} \cdot \vec{n}$

Finite volume method

The vector form of the Euler equations holds for any control volume (change of mass, momentum and energy is equal to the fluxes through the boundaries of the volume).

To get a discretization of the equations, we need to create a mesh with cells being multiple small control volumes Ω.

Finite volume method

Assuming $\vec{W}$ to be constant over the control volume Ω:

$$\int_{\Omega} \frac{\partial \vec{W}}{\partial t} = \Omega \frac{\partial \vec{W}}{\partial t}$$

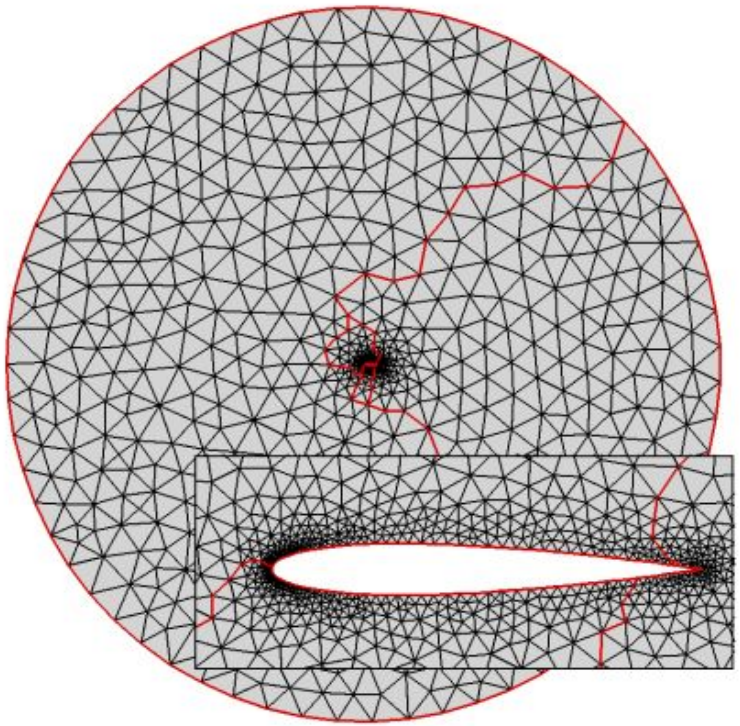
Taking the control volumes to have N_F planar faces and assuming the fluxes to be constant over the face:

$$\oint_S \vec{F_c} \, \mathrm{d}S = \sum_{m=1}^{N_F} (\vec{F_c})_m S_m = \vec{R}$$

Steps of a CFD solver

1. Read the mesh
2. Compute the geometrical quantities (cell volume, face surface, face normal, etc.)
3. Set the initial and boundary conditions
4. Iterate over time (until convergence or desired time):
    a. Loop on faces to compute the fluxes;
    b. Compute the residual $\vec{R}$ of each cell;
    c. Advance $\vec{W}$ in time (explicitly or implicitly);
    d. Update the boundary conditions;
    e. Compute the forces and moments (integrate the pressure at the surface);
    f. Monitor convergence
5. Post process (write the field solution, the solid wall solution, the force and moment coefficients).

2D Mesh – Multiblock unstructured

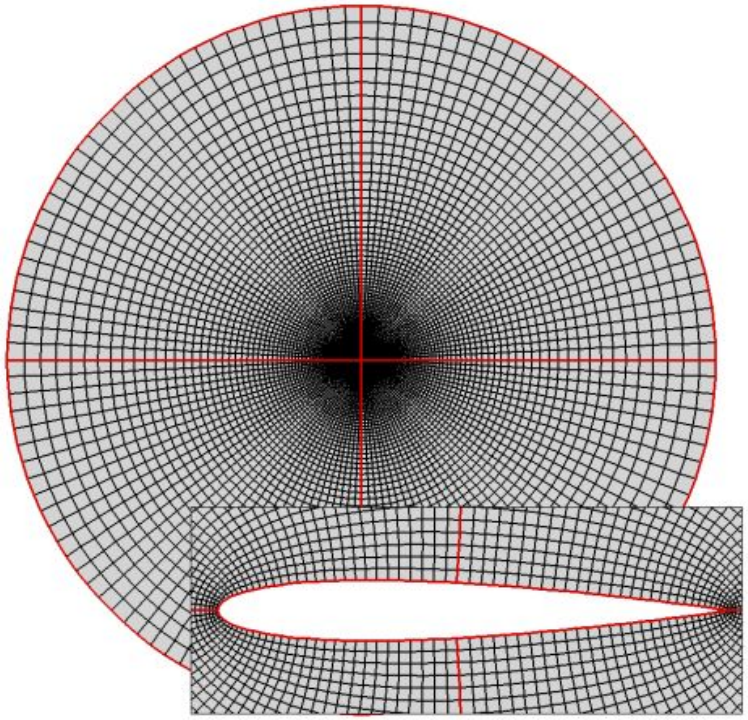


Red lines are block boundaries

- A node can be connected to an arbitrary number of cells;
- Cells can have an arbitrary number of faces;
- The connectivity is explicit;
- Loops are done on number of nodes, number of faces and number of elements.

Not used in this course

2D Mesh – Multiblock structured



Red lines are block boundaries

- Every node connects 4 elements;
- Cells are quads;
- The mesh connectivity is implicit;
- A mesh cell can be identified with an i, j index;
- Mesh can have an arbitrary number of blocks, but all are $n_i \times n_j$;
- Loops on elements, nodes and faces are ordered $(n_i \times n_j)$

In Homework 2, we simplify by using single block structured grids

Why choose a mesh type?

Structured:

- Less memory usage;
- Computationally more efficient;
- Numerically more precise (if grid quality is good);
- Grids are of better quality;
- All loops are ordered in i, j (and k);
- Powerful acceleration techniques available (multigrid is very effective).

Cons: good quality grids are hard to produce (especially in 3D)

Unstructured:

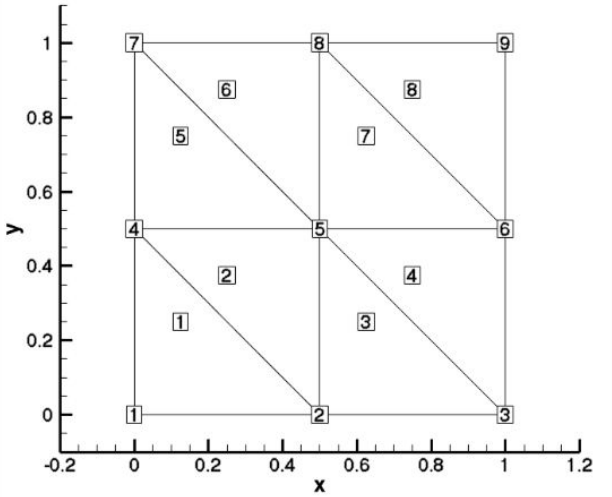
- Grids are easier to produce;
- Source code can be "clearer" (loops on faces, cells or nodes are explicit);
- Multigrid methods are harder to implement and less effective.

Unstructured solvers are more and more popular because of the simplicity of grid generation.

Unstructured mesh file (SU2)

Not used in this course

1-based



1. We need the coordinates of the nodes;
2. Connectivity list of the elements to nodes;

Coordinates

```
NDIME= 2
NPOIN= 9
0.00000000000000 0.00000000000000
0.50000000000000 0.00000000000000
1.00000000000000 0.00000000000000
0.00000000000000 0.50000000000000
0.50000000000000 0.50000000000000
1.00000000000000 0.50000000000000
0.00000000000000 1.00000000000000
0.50000000000000 1.00000000000000
1.00000000000000 1.00000000000000
```

```
NELEM= 8
5 0 1 3
5 1 4 3
5 1 2 4
5 2 5 4
5 3 4 6
5 4 7 6
5 4 5 7
5 5 8 7
```

Element list (0-based)

Unstructured mesh file (SU2)

Not used in this course

We need a standard for how the connectivity from element to node is done. For example the VTK one:
`https://vtk.org/wp-content/uploads/2015/04/file-formats.pdf`

Element Type	Identifier
Line	3
Triangle	5
Quadrilateral	9
Tetrahedral	10
Hexahedral	12
Prism	13
Pyramid	14

Unstructured mesh file (SU2)

Not used in this course

We also need to define the boundary conditions

```
NMARK= 4
MARKER_TAG= lower
MARKER_ELEMS= 2
3 0 1
3 1 2
MARKER_TAG= right
MARKER_ELEMS= 2
3 2 5
3 5 8
MARKER_TAG= upper
MARKER_ELEMS= 2
3 8 7
3 7 6
MARKER_TAG= left
MARKER_ELEMS= 2
3 6 3
3 3 0
```

This is the list of lines forming the different BC of the domain. A type must be attributed to these markers elsewhere.

Connectivity tables

Not used in this course

In the solver, we need:

- List of elements touching a node;
- List of elements connected to a face;
- etc.

There exists an efficient algorithm to construct these tables:

Lohner, R. (2008). *Applied computational fluid dynamics techniques: an introduction based on finite element methods*. John Wiley & Sons.

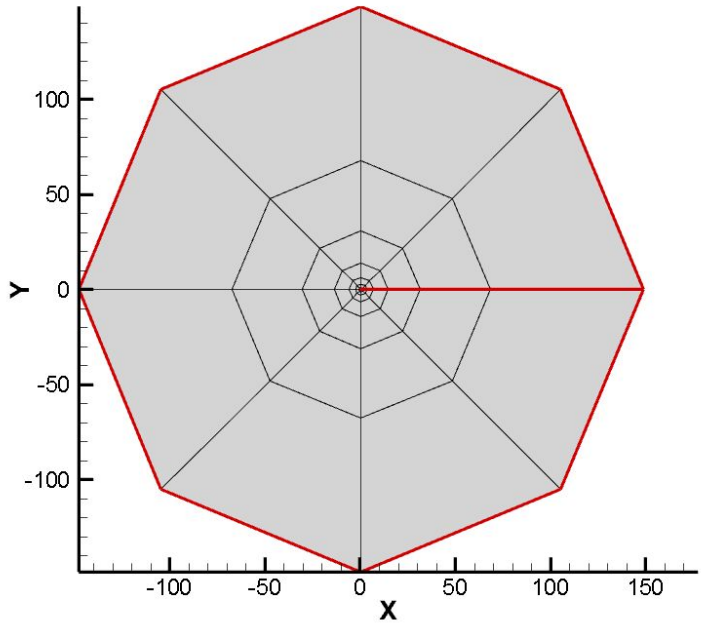
Structured plot3D file format

Meshes are provided on Moodle in plot3D format:

- An index for the number of blocks m;
- m sets of 3 indexes n_i n_j;
- A list of coordinates in the order:
  - All x of block 1 (order i, j, k);
  - All y of block 1 (order i, j, k);
  - All x of block 2 (order i, j, k);
  - etc.

Meshes are provided from 9×9 to 2049×2049.

Add 2 layers of ghost cells for the boundary conditions implementation



Structured plot3D file format

We would need a second file to describe the boundary conditions:

- Wall;
- Farfield;
- Connection.

In the homework, you can hard code this:

- Farfield is j_{max};
- Wall is j_{min};
- i_{min} is connected to i_{max}.

Examples of Tecplot ASCII files

```
TITLE="NODE_CENTERED"
VARIABLES = "X", "Y","P"
ZONE I=3, J=3,
datapacking=block,
zonetype=ORDERED,
varlocation=([1,2,3]=nodal)
0,1,2,0,1,2,0,1,2
0,0,0,1,1,1,2,2,2
1,2,3,4,5,6,7,8,9
```

Mesh sizes / structured data; x values, y values, then P (by block, 9 points)

```
TITLE="NODE_CENTERED"
VARIABLES = "X", "Y","P"
ZONE I=3, J=3,
datapacking=point,
zonetype=ORDERED,
varlocation=([1,2,3]=nodal)
0, 0, 1
1, 0, 2
2, 0, 3
0, 1, 4
1, 1, 5
2, 1, 6
0, 2, 7
1, 2, 8
2, 2, 9
```

Trios for x, y, P (point format); the comma can be replaced by spaces

In Linux, the command `preplot` can be used in a terminal to convert an ASCII file to binary. Binary files take less space and are faster to open. This can be useful for large meshes.

Other formats are compatible with Tecplot (like plot3D files).

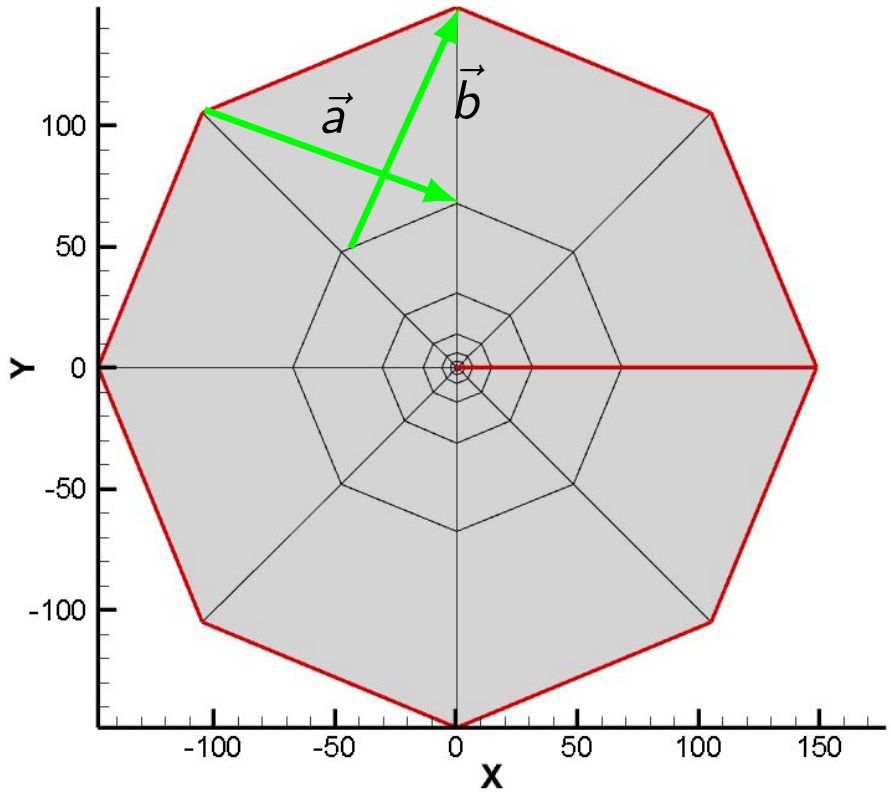
Interpolate the variables from cell-centre to the node when writing them.

Computation of mesh metrics

volume $= 0.5\,|\vec{a} \times \vec{b}|$
(volume in 2D is an area)

Normal vector is the transpose of the edge vector. Store the normalized vector. Also store the face area (length in 2D).

Note: Be careful about the normal vector direction. It needs to be consistent. Identify a "left" and "right" cell with respect to the face and define the normal from "left" to "right". Left and right in a topological sense.



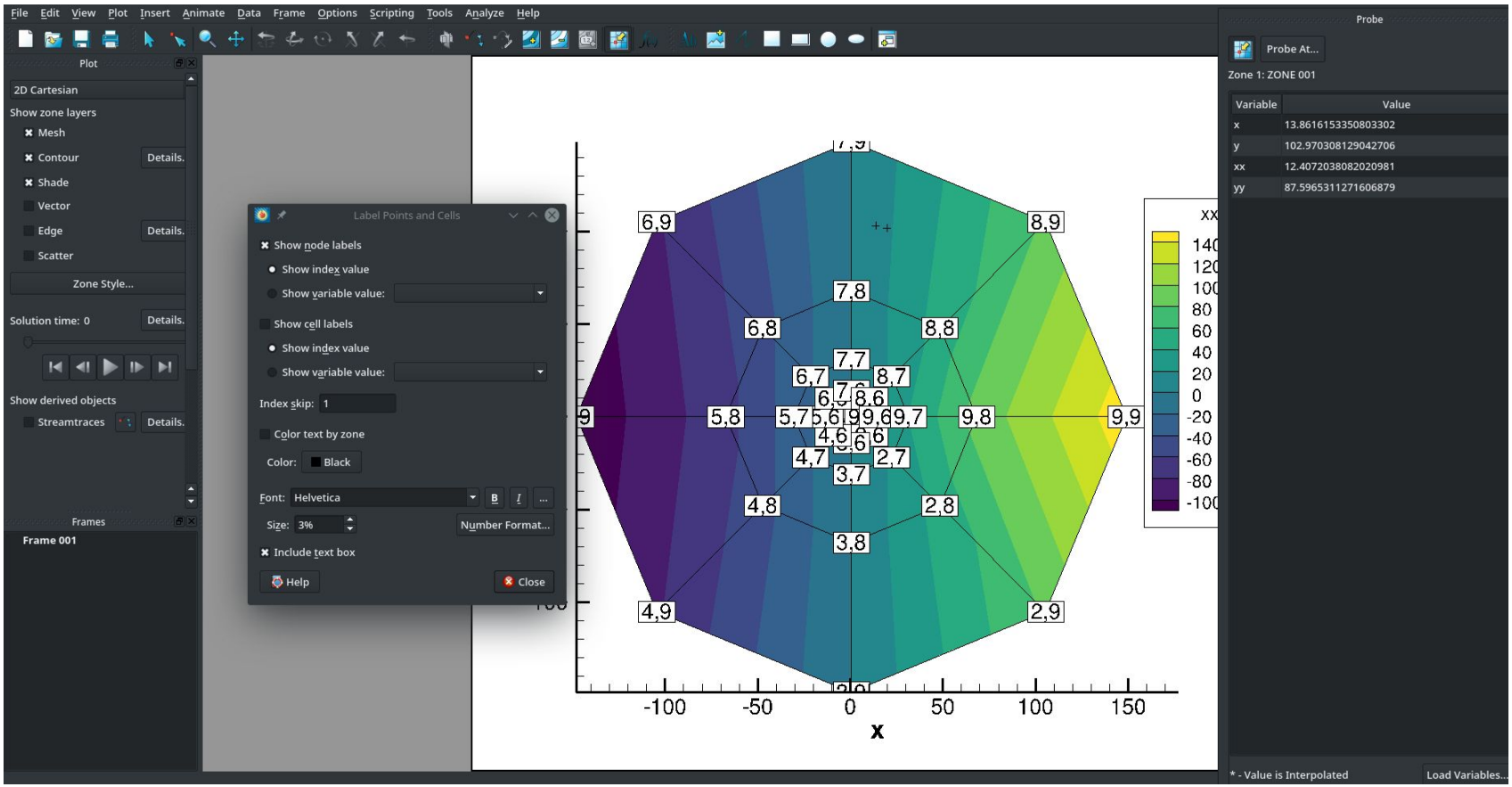
Homework 2 – first steps

Note that an example code with empty structure will be provided.

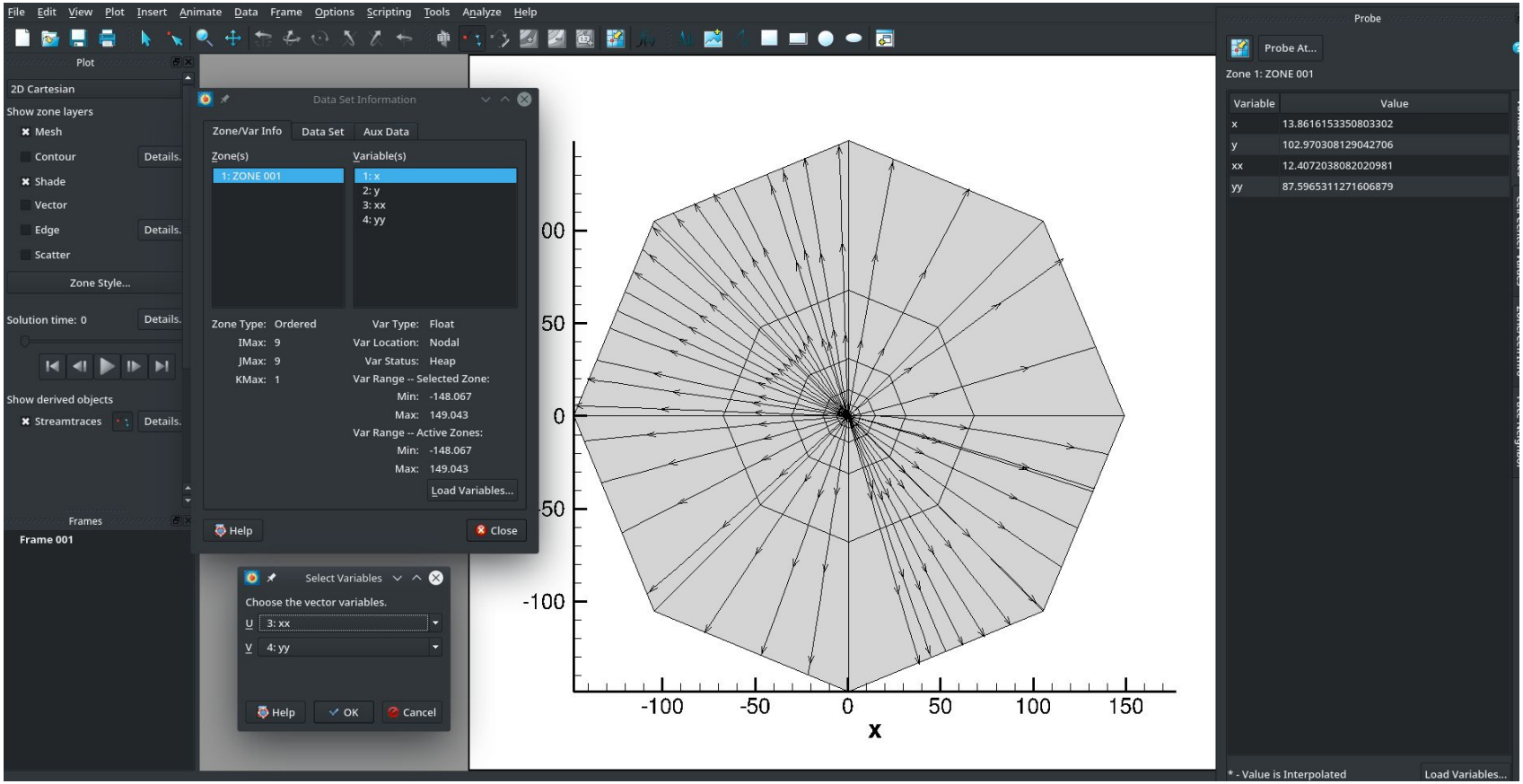
1. Prepare a data structure;
2. Read the mesh file;
3. Compute and store the mesh cell volumes and face lengths;
4. Write back the mesh with the mesh cell volumes;
5. Verify the metrics by setting the flux ($\vec{F}_c$) equal to 1. Then the residual must be 0 (the integral of the normal over the surface of a closed volume is 0).

When debugging, try first on a Cartesian mesh to have normals aligned in the x and y directions. Write the mesh on paper with node, face and cell indices and print in the code to verify your connectivity and normal directions.

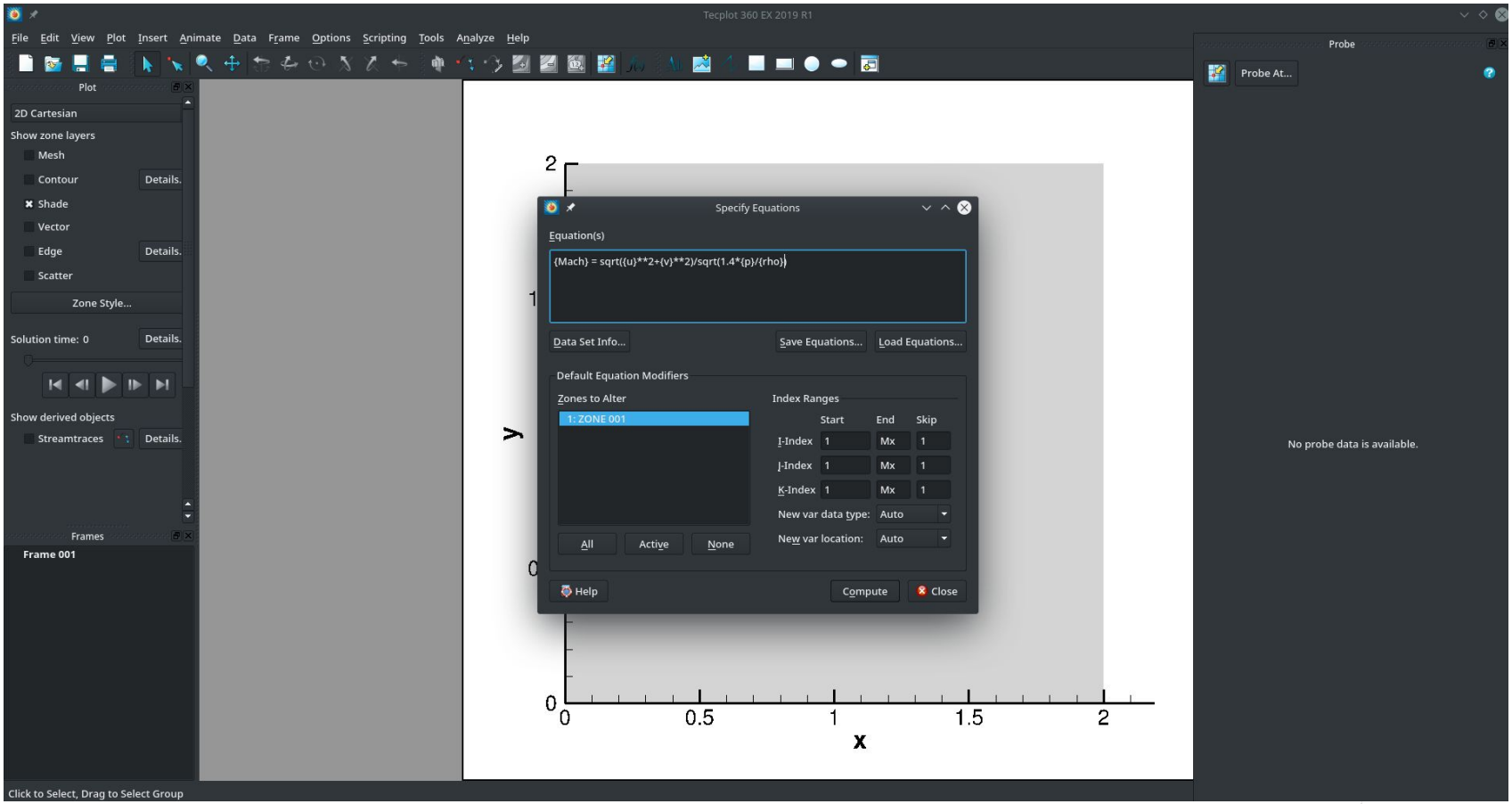
Tecplot introduction



Tecplot introduction



Tecplot introduction



Example script: cell centre coordinates

